RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Degree, Semester & Branch: VII Semester B.E. ECE B
Course Code & Title: EC6703 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar
Name of the Topic: Cache Memory
Name of the Innovative Practice: Role Play
Date & Duration: 15.07.2019 & 20 minutes
ICT Tool Used: LCD projector, Personal Computer
Description:

Role-play is a technique that allows students to expose different view points or ways of thinking about a situation, expand their ability to resolve situations, and provide experience within a given context.

Goals (Learning Outcomes):

- The students will able understand the concept Main Memory and Cache Memory.

Use of appropriate methods:

Justification for choosing activity:

Cache memory, also called CPU memory, is high-speed static random access memory (SRAM) that a computer microprocessor can access more quickly than it can access regular random access memory (RAM). This memory is typically integrated directly into the CPU chip or placed on a separate chip that has a separate bus interconnect with the CPU.

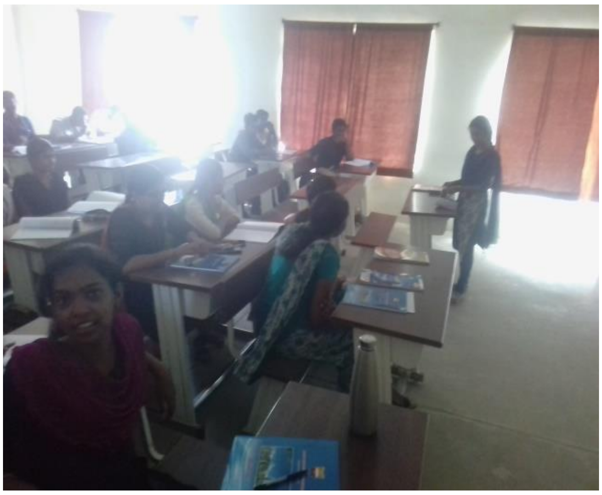
The purpose of cache memory is to store program instructions and data that are used repeatedly in the operation of programs or information that the CPU is likely to need next. The computer processor can access this information quickly from the cache rather than having to get it from computer's main memory. Fast access to these instructions increases the overall speed of the program. It is very important topic , students should have much exposure about cache memory.

Effective presentation

Details of the Implementation:

In this two containers are used,container one is considered as main memory, container two is considered as cache memory.1 to 50 numbered tokens are placed in container one and few token from 1 to 50 placed in container two.Through that explained cache hit, cache miss concept and analysed time requirement of cache hit and cache miss operation.

Innovative Teaching Method Execution
Cache Memory - Role Play

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated at end of role play, Most of the students views about the topics are valid and they understand the use of cache memory.

Reflective Critique:

Benefits:

- Role play increases the understanding of topic, they easily remember during exams.
- Generates more ideas about application of cache memory.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	-	-	-	2	-	2	-	-

CO1: The Students will be able to explain the fundamental concepts of designing Embedded and Real Time Systems.

Reference:

- Marilyn Wolf, "Computers as Components - Principles of Embedded Computing System Design", Third Edition "Morgan Kaufmann Publisher,2012.
- https://computersciencewiki.org/index.php/Cache_memory
- https://nptel.ac.in/courses/106102062/

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Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title:EC6703 Embedded and Real Time Systems
Name of the Faculty member:Mr.G.Sivakumar
Name of the Topic: DMA(Direct Memory Access)
Name of the Innovative Practice: Quiz
Date & Time: 18.07.2019& 15 Minutes
ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

The Quiz activity module allows the teacher to design and build quizzes consisting of a large variety of Question types, including multiple choice, true-false, short answer and drag and drop images and text. These questions are kept in the Question bank and can be re-used in different quizzes.

Goals (Learning Outcomes):

- The students will learn and understand the concepts data transfer schemes and how DMA transfer data between memory and I/O devices.

Use of appropriate methods:

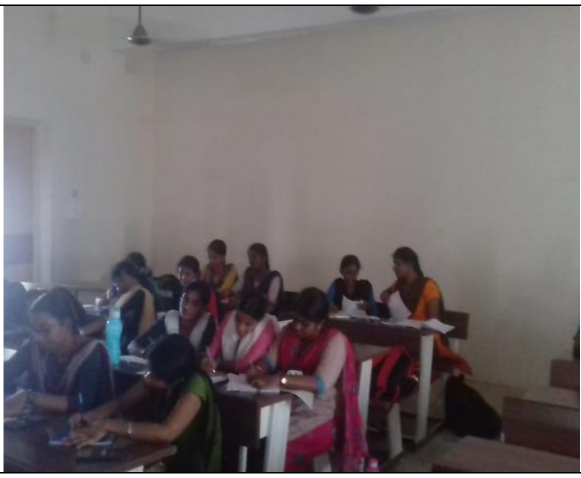
Justification for choosing the activity:

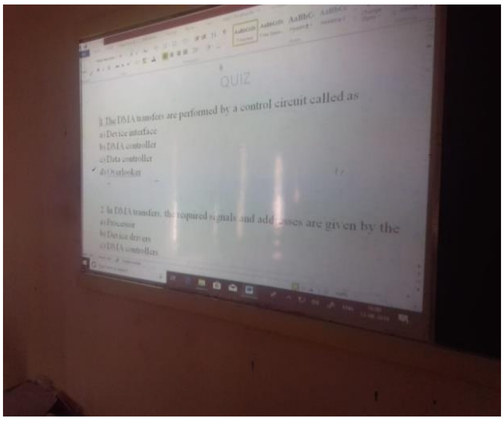
The data transfer take place between microprocessor and memory, microprocessor and I/O devices and memory & I/O devices.There are three types of data transfer schemes used to transfer data between microprocessor and memory. The functionalities of those data transfer schemes differ widely and hence this quiz with multiple choices helps the students to think in multiple dimensions and provide the answers.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution
DMA (Direct Memory Access)- Quiz





Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- Most of the questions were correctly answered by the students. The wrongly answered questions were discussed at the end of the class.

Reflective Critique:

Benefits:

- The quiz was conducted with multiple choices. The choices that were given were closely related to each other which made the students to think in a deeper manner and give the answers.

Challenges:

- The time duration was exceeded as some of the students took longer time to answer some questions.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	2	1	-	-	-	2	-	2	-	-

CO2: The Students will be able to describe the computing required for Real Time Embedded Systems.

Reference:

- Marilyn Wolf, "Computers as Components - Principles of Embedded Computing System Design", Third Edition "Morgan Kaufmann Publisher,2012.

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<u>Innovative Practices Description</u>

Degree, Semester & Branch: VII Semester B.E. ECE B
Course Code & Title: EC6703 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar
Name of the Topic: Multitask and Multiple Process
Name of the Innovative Practice: Demonstration
Date & Duration: 06.08.2019 & 15 Minutes
ICT Tool Used: LCD projector, Personal Computer
Description:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. In demonstration method, the teaching-learning process is carried in a systematic way. Demonstration often occurs when students are unable to understand theories concept.

Demonstrations help to raise student interest and reinforce memory retention because they provide connections between facts and real-world applications of those facts. Lectures, on the other hand, are often geared more towards factual presentation than connective learning.

<u>Goals (Learning Outcomes):</u>

Learning Outcomes:

- The students will able understand the concept and difference between the program, process, task, thread.

<u>Use of appropriate methods:</u>

Justification for choosing Demonstration activity:

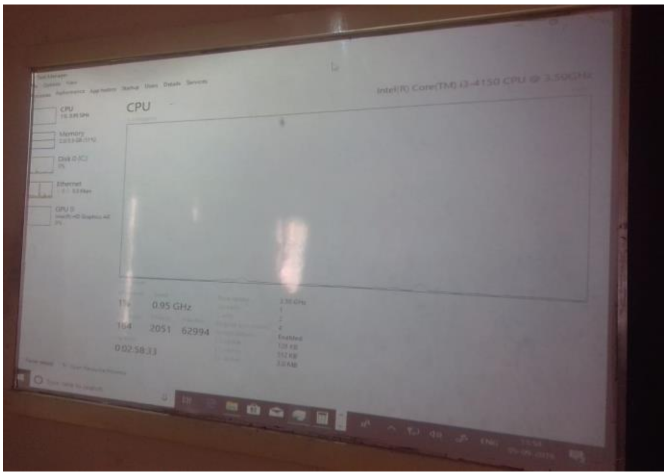
The task, process and thread are important concepts in real time operating system, student known the definition of task, process and thread but they were not able to differentiate. Through live demonstration students will able to know the above concepts.

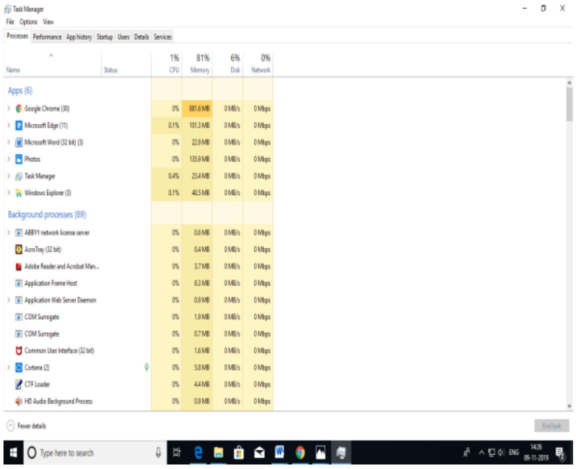
<u>Effective presentation</u>

Details of the Implementation:

While teaching the topic of process and operating system, students got confusion in program, task, process, thread, multitask, multiprocessing and multiprogramming. So explained these concepts using task manager in personal computer. Task manager shows information like CPU, Memory utilization, number of task sharing the CPU and number of threads executing in CPU.

Innovative Teaching Method Execution
Multitask and Multiple Process - Demonstration





Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- While the demo was done live by opening the Task Manager in the personal computer and showing the task, process and thread in it, students were very eager to watch they understood the difference in a very better manner.

Reflective Critique:

- **Benefits:**
  - After completion of the demonstration, all the students were able to differentiate the concept program, task, process, thread, multitask, multiprocessing and multiprogramming.

- **Challenges:**
  - Some of the students found difficult to follow the content, so demonstrated again with examples, it took some extra time to finish the demonstration**.**

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	2	2	2	3	-	-	1	-	1	-	1

CO3: The Students will be able to summarize the concepts of scheduling in Real Time Operating System

References:

- Marilyn Wolf, "Computers as Components - Principles of Embedded Computing System Design", Third Edition "Morgan Kaufmann Publisher,2012.
- https://www.udacity.com/course/ud923

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Innovative Practices Description

Degree, Semester & Branch: VII Semester B.E. ECE B
Course Code & Title: EC6703 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar
Name of the Topic: Design Methodologies
Name of the Innovative Practice: Minute paper
Date & Duration: 28.08.2019 & 20 Minutes
ICT Tool Used: LCD projector, Personal Computer

Description:

A "minute paper" may be defined as a very short, in-class writing in response to an instructor-posed question. Its major advantage is that it provides rapid feedback on whether the professor's main idea and what the students perceived as the main idea are the same.

Additionally, by asking students to add a question at the end, this assessment becomes an integrative task. The purpose of the one-minute paper was to assess student learning at the end of a day's lesson.

Goals (Learning Outcomes):

Learning Outcomes:

- The students can easily differentiate the different types of design methodologies and will be able to list the pros and cons of different design methodologies.

Use of appropriate methods:

Justification for choosing minute activity:

Minute papers have been found to be effective in providing a conceptual bridge as it bridges the gap between theory and practical knowledge. The design methodologies have five different designing methods, the minute paper activity helps students to increase the attentiveness and recollect the concepts.

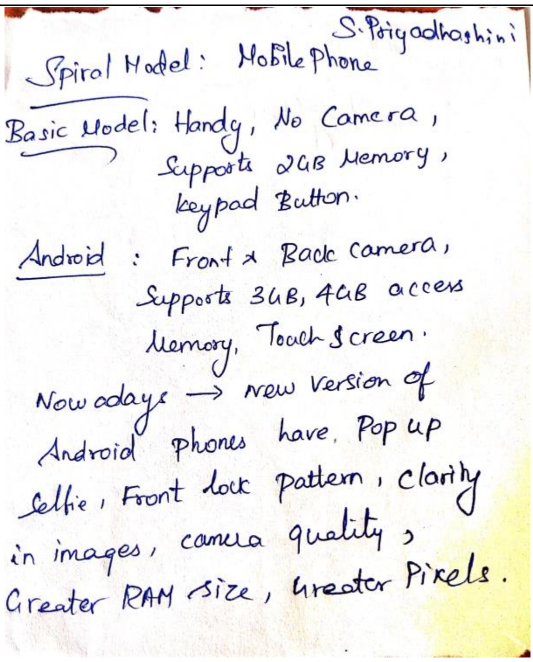
Effective presentation

Details of the Implementation:

After completion of topic, asked few questions to students, then came to know some they got confusion because Design methodologies has five types of design flow, each of that used for different applications. The Students are insisted about mind map activity and asked to go through the topic, next class students are asked to write answer for given questions.

Innovative Teaching Method Execution

Design Methodologies - Minute paper

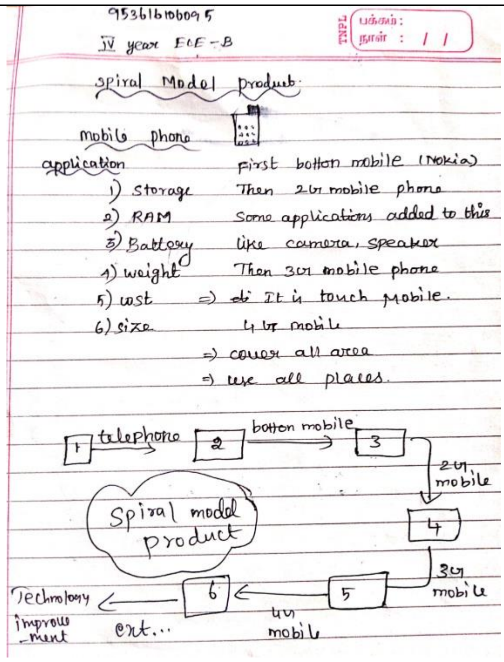

S.Priyadharshini

Spiral Model: Mobile Phone

Basic Model: Handy, No Camera, Supports 2GB Memory, keypad Button.

Android : Front & Back camera, Supports 3GB, 4GB access Memory, Touch Screen.

Now adays → new Version of Android phones have, Pop up Selfie, Front lock pattern, clarity in images, camera quality, Greater RAM size, Greater Pixels.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IV year ECE-B

spiral Model product

mobile phone application

1) storage

2) RAM

3) Battery

4) weight

5) cost

6) size

First botton mobile (Nokia)

Then 2G mobile phone

Some applications added to this like camera, speaker

Then 3G mobile phone

=> It is touch mobile.

4G mobile

=> cover all area

=> use all places.

<u>Significance of Results:</u>

Assessment of Effectiveness/Success of the Activity:

- Most of the students written correct answer in minute paper.
- The success of the activity was evaluated by asking the same question in Internal Assessment test III, more than 80% of students attended and answered correctly.

<u>Reflective Critique:</u>

Benefits:

- Design methodologies has five types of design flow, in minute paper the question is which design flow is suitable to produce different version of electronics products. The students felt easy to remember and recollect the concepts of different types of design flows and they written answer.

Challenges:

- A day before informed about the activity and topic, but students asked 15 minutes for reviews the topic. I have planned minute paper activity for 2 minutes but it took 20 minutes.

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO4	3	2	3	1	-	-	-	2	-	2	-	2

CO4: The students will be able to analyze the techniques required to create complex Embedded system.

Reference:

- Wayne Wolf, "Computers as Components - Principles of Embedded Computer System Design", Morgan Kaufmann, 2nd Edition, 2008.
- http://vidyamitra.inflibnet.ac.in/index.php/search?subject%5B%5D=&course%5B%5D=Embedded+system&domain%5B%5D=Engineering+and+Technology
- https://provost.tufts.edu/celt/files/MinutePaper.pdf

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Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6703 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar
Name of the Topic: Digital Camera
Name of the Innovative Practice: Animated Videos
Date & Time: 27.09.2019 & 15 Minutes
ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

Learning Outcomes:

- The students can acquire better knowledge about the operation digital Camera.

Use of appropriate methods:

Justification for choosing Animated Videos:

The students often have a difficulty to understand exactly how a digital camera works, this video clip clearly explain the operation of digital camera, the biggest benefits of this animated video is, it describes the complex operation of image formation and shutter operation in a simple way.

Effective presentation:

Details of the Implementation:

While handling the topic digital camera in the class hour, students were under confusion regarding the working operation of digital camera. Then the video was shown and almost all the students were able to understand the concept.

Innovative Teaching Method Execution

Digital Camera - Animated Videos	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III most of students attended and answered correctly.

Reflective Critique:

- **Benefits:**

  The students were more interested in learning the concepts if we visually show the videos. They will be able to easily understand working of digital camera and they will easily remember during exams.

- **Challenges:**

  It takes some time to choose relevant videos for working of digital camera.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	1	2	2	1	-	-	1	-	1	-	1

CO5: The students will be able to apply the concepts of Embedded System Design in creating the model Real Time applications

Reference :

- David. E. Simon, "An Embedded Software Primer", 1st Edition, Fifth Impression, Addison Wesley Professional, 2007.
- https://www.khanacademy.org/science/electrical-engineering/reverse-engin/digital-camera/v/what-is-inside-a-digital-camera-1-of-2-1
- https://fstoppers.com/photography-101-how-to-use-your-digital-camera-and-edit-photos-in-photoshop